

Complex Segments and the Perceptual Compression Hypothesis

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Abstract

This proposal introduces a new principle, the Perceptual Compression Hypothesis (PCH), which postulates that subsyllabic elements—onset, nucleus, and coda—represent segmental temporal units in phonetics capable of hosting multiple phonetic segments, thereby giving rise to complex segments. Following the tenets of Strict CV theory, which proposes that CV is the only syllable structure existing in phonology, this framework argues that the onset position admits only complex segments, while the arrangement of complex segments within rhyme elements depends on grammatical patterns exclusive to each language.

The hypothesis distinguishes between two types of complex segments: primary complex segments, which obtain their linear order from a subsegmental algorithm following Wagner Oviedo (2026), and derived complex segments, which occur during derivation through the fusion of linear elements into a single segmental temporal position. To identify these structures, three diagnostics are proposed: predictability, separability, and distributional identification.

The proposal further argues that, at a specific point in the derivation, primary and derived complex segments neutralize, exhibiting identical effects in phonological processes. Finally,

this model is applied to three languages showing distinct arrangements: Spanish, characterized by more derived than primary complex segments; Sanskrit, with a similar distribution of both; and Huautla Mazatec, which features a predominance of primary complex segments.

1.- Introduction

I propose that the syllable onset position admits only a single prosodic slot of type C. This stands in contrast to the rhyme, which may host multiple prosodic elements of type V or C. This proposal extends the framework established by Duanmu (2009), which assumes that all onset elements behave as complex segments.

However, this study justifies a formal distinction between two specific types of segmental complexity. Primitive complex segments are defined as those whose linearization is governed by internal or sub-phonemic rules. In contrast, derived complex segments are those whose linearization is determined by the inherent lexical order of phonemes, yet their phonetic expression is realized within a single C or V prosodic position—or across multiple positions if the internal melodic specifications are identical.

The onset is defined as the initial consonantal margin of the syllable—a phonological unit that, according to Zec (2007), represents "phonologically significant groupings of segments." Although the exact delimitation of the syllable varies across languages, we can follow Hayes

(2009) in recognizing it as a unit formed around a nucleus, which is identified as the most sonorous segment in a given sequence.

This follows the Sonority Sequencing Principle (Clements, 1990; Selkirk, 1984; Zec, 2007), which establishes that syllables tend to group their constituents around the most sonorous segment at their core. Consequently, the margins—the onset and the coda—are recognized as less sonorous than the nucleus. In the onset position, segments are preferentially ordered in a sequence of rising sonority toward the nucleus, whereas the inverse is true for codas, which typically exhibit falling sonority.

Despite the theoretical elegance of the Sonority Sequencing Principle (SSP), numerous apparent counter-examples exist. In onset positions, certain sequences notoriously violate the ideal rising sonority profile—most notably fricative-plus-plosive (sC) sequences and prenasalized stops (nC). In these cases, the initial element is more sonorous than the subsequent one, creating a sonority reversal.

To preserve the SSP, several theoretical solutions have been proposed. One approach is to assume these sequences are heterosyllabic (s.C or n.C), thereby removing the violation from the onset constituent itself. An alternative, not necessarily mutually exclusive, is to treat these sequences as complex segments. Under this view, the sonority violation is merely a phonetic surface reality, while phonologically, the syllable maintains a standard CV shape.

Notable among these "complex segment" solutions is Steriade's (1994) Aperture Theory, which analyzes sonority-violating sequences in Huautla Mazatec. Steriade argues that surface sequences like [nkʔ] are underlyingly non-complex /nkʔ/ strings that emerge as

complex in the phonetics. However, this framework treats sonority primarily as a surface phenomenon.

Departing from this, Wagner Oviedo (2026) considers that sonority is a phonological primitive following Scheer (2022). In this view, sequences like [nkʔ] are complex segments in both the phonetics and the phonology, with their linearization resulting from sub-phonemic rules within a hierarchical, multi-layered structure.

Furthermore, Duanmu (2009) proposes a strict CVX syllable weight constraint, suggesting that sequences like /kl/ or /pr/ are complex segments occupying a single C slot. Building on these foundations, this work identifies two levels of segmental complexity underlying (following Wagner Oviedo, 2026) and surface. We adopt Duanmu's (2009) premise that the onset position consists persistently of only a single C slot—a principle we term "uncomplex onsets"—while distinguishing between underlying and surface manifestations.

Underlying onsets exist in the lexical representation, whereas surface onsets emerge through derivation, as Steriade (1994) suggests. To maintain the requirement that an onset occupy only one slot, we distinguish between two types: primitive and derived complex segments. Primitive complex segments occupy the onset position both underlyingly and on the surface, while derived complex segments occupy the onset position only at the surface level. While these two types may appear indistinguishable on the surface, they can be differentiated through three specific parameters: predictability, separability, and distribution.

The predictability parameter, for instance, determines whether a complex phoneme can be traced through morphological analysis. In Mixe (Herrera Zendejas, 2009), creaky and breathy nasals appear only in morphologically derived forms, signalling their status as derived

segments. Conversely, in Mazatecan languages (Golston & Kehrein, 1998; Wagner Oviedo, 2026), such nasals appear primarily in non-derived roots. This lack of predictability identifies them as primitive complex segments.

Predictability also manifests through the allophonic realization of a phoneme as a surface-level complex segment during the derivation process. For example, in Ixcatlán Mazatec (Wagner Oviedo, 2018)), the sequence [h^w] appears exclusively before /a/, whereas [ϕ] occurs before /e/, /i/, and /o/. This indicates a complementary distribution in which the complex segment is an allophone of the phoneme /ϕ/.

Diachronically, the phoneme /ϕ/ originated from the historical coalescence of /hw/ sequences. Consequently, the contemporary allophonic variation allows us to trace the complex segment as a remnant of the original sequence's manifestation. This reinforces the idea that surface complexity can be a predictable outcome of both internal phonological rules and historical sound change.

The separability parameter refers to whether a sequence can be split during syllabification or through native speaker identification of syllable boundaries (Trubetzkoy, 1939). For example, in Ixcatlán Mazatec, the prenasalized /ⁿg/ and the affricates /ts/ and /tʃ/ are analyzed as complex phonemes (Wagner Oviedo, 2018). Consequently, in the syllabification of words such as /ma².ⁿga²/ ‘he runs’, /t^ho².tso²/ ‘canoe’, and /ki.tʃa/ ‘metal’, the constituents of these complex surface phonemes are never separated.

In contrast, Mexican Spanish treats only /tʃ/ in this manner. In words like /manga/ ‘sleeve’, /pitsa/ ‘pizza’, and /pitʃa/ ‘he invites’, the sequences are split in the first two cases—/man.ga/ and /pit.sa/—whereas only the affricate remains intact: /pi.tʃa/. This indicates that in Mexican

Spanish, only /tʃ/ functions as an underlyingly complex segment, while /n.g/ and /t.s/ are neither underlying nor derived complex segments. In Mazatec, however, all three sequences are underlyingly complex.

The separability parameter is even more robust when morphologically conditioned alternations break these sequences and parse their constituents into different syllables. As discussed in Section 3, Mexican Spanish utilizes diphthongs as derived complex segments. In both inflectional and derivational processes, diphthongs that occupy a single slot in one context are split and redistributed into separate syllables in another. For instance, the verb /basiar/ ‘to empty’ is syllabified as /ba.siar/ in its infinitive form; however, the third-person present indicative form is syllabified as /ba.si.a/. This "breaking" of the /ia/ diphthong under morphological pressure clearly demonstrates its status as a derived segment rather than a primitive one.

The final parameter is distribution. While considered the least important diagnostic of the three, it presupposes that underlyingly complex segments tend to exhibit systemic symmetry (Trubetzkoy, 1939), correlating with non-complex series within the phonological system. For example, in Mazatecan languages, the prenasalized series /ⁿd, ⁿdz, ⁿdʒ, ⁿg/ forms a direct correlation with the simple plosive series /t, ts, tʃ, k/.

Paradoxically, the distribution parameter also suggests that underlyingly complex segments often show a more restricted distribution compared to derived complex segments. In Mexican Spanish, for instance, sequences of obstruents plus liquids are largely unconstrained, appearing in a wide array of combinations: /p, tʎ, kʎ, pr, tr, kr, bl, gl, br, dr, gr. The only notable exceptions are the prohibited sequence /dl/ and those involving strident sounds (/sl, sr, tʃl, tʃr/), which are invariably heterosyllabic, as seen in /is.la/ ‘island’ and /is.ra.el/ ‘Israel’.

This lack of symmetry is further evidenced by the absence of other potential complex segments; while /tʃ/ is a stable part of the inventory, there is a total inexistence of corresponding pairs such as /pʃ/ or /kʃ/. This distributional isolation supports the status of /tʃ/ as a singular primitive complex segment in the language, as it lacks the broader systemic correlations typical of more expansive complex series. While these three parameters serve as a diagnostic to differentiate between primitive and derived complex segments, the motivation for the uncomplex onsets proposal requires justification

It is a well-established cross-linguistic fact that the CV shape is the most common syllable structure in human languages (Easterday, 2019; Gordon, 2016; Jakobson, 1941; Steriade, 1994). In fact, proponents of the Strict CV model (Faust & Ulfsgjorninn, 2025; Lowenstamm, 1996; Scheer, 2004), argue that this is the only possible syllable type in phonology.

Gordon (2016) provides a perceptual basis for this CV preference, noting that the speech signal benefits from alternating periods of low and high intensity. He explains:

“Another perceptual advantage conferred upon CV syllables relates to the virtues of having a speech signal that consists of alternating periods of low intensity, the consonants, and high intensity, the vowels. These modulations maximize the firing rates of auditory nerve fibers (and perceived loudness), which display the greatest response to a sound at its onset before adaptation gradually reduces sensitivity to a sustained stimulus. A period of low intensity, as during an onset consonant, affords the auditory system a chance to recover so that it is maximally responsive to the next phase of relatively high intensity in the speech signal. Auditory recovery offers an

explanation for the cross-linguistic preference for lower sonority onsets, which, because of their reduced intensity, enhance the response of the auditory system to the following vowel.”

This explanation suggests an intrinsically motivated relationship between low intensity and maximal auditory responsiveness in the onset, versus high intensity and reduced sensitivity (due to adaptation) in the rhyme. However, these facts are inversely correlated with the observation that the rhyme is the primary locus for phonological duration perception, whereas onsets typically do not license such contrasts—though languages like Ogami (Pellard, 2010) or Valley Zapotec (Arellanes, 2009; Chávez Peón, 2010) are notable for displaying "initial geminates" (Topintzi, 2008; Topintzi & Davis, 2017).

There is a particular relationship between the perception of intensity and the perception of duration in the cognitive realm, as described over a century ago by Bolton (1894) and Woodrow (1909). Specifically, Woodrow (1909) demonstrates that when a stronger stimulus precedes a temporal interval, that interval is perceived as longer. While Hayes (1995) utilizes this principle to propose the Iambic-Trochaic Law, we can extend this logic to the CV syllabic profile.

Although the C position is physically less intense than the V position—which, following Woodrow, should theoretically make the subsequent interval seem longer—the reality of speech perception is governed by the "auditory recovery" described by Gordon (2016). Because the C position coincides with the most optimal firing response, internal intensity perception aligns itself with this external low-edge value. The onset thus represents the point

of departure from which the following V phase is perceived as a measurable interval. This makes the rhyme uniquely amenable to temporal segmentation and discrimination, whether in the realm of duration (short vs. long) or melody (segment x vs. y).

These facts explain not only the cross-linguistic tendency to avoid duration contrasts in the onset, but also why the onset licenses a denser array of melodic contrasts. In the lexicon of Mexican Spanish, for example, consonants in the coda are largely restricted to coronals, with other places of articulation being highly limited or entirely non-existent for many speakers. In contrast, the onset position remains the primary site for maximal phonemic diversity.

Regarding uncomplex onsets, we can predict that the C position naturally tends to aggregate greater segmental distinctions over time, including the formation of complex segments. The reinterpretation of phonetic sequences—such as [nkʔ] in Mazatecan languages (Chávez Peón Herrero & Filio García, 2021; Chávez Peón et al., 2025; Pike & Pike, 1947; Wagner Oviedo, 2026) or complex clicks like [lkχʔ] in !Xóõ (Traill, 2018)—as unitary complex segments can be explained by this maximal perception edge.

To formalize this mechanism, I propose the Perceptual Compression Hypothesis (PCH). The PCH posits that because the onset coincides with the most optimal firing response of the auditory system, the cognitive system prioritizes the integration of melodic features over the linear ordering of segments. Within this "maximal perception edge," heightened auditory responsiveness causes rapid sequences to be perceived as a single temporal entity rather than a succession of discrete phonological events.

Because the onset acts as the perceptual "trigger" for the syllable, the listener's auditory system effectively "compresses" the melodic bundle. This perceptual compression provides

the functional grounds for phonological mergers: historically distinct segments are reanalyzed as a single complex phoneme occupying the sole available C slot.

This mechanism ensures that even as the melodic complexity of a language increases, the fundamental "uncomplex" CV prosodic template remains intact. Under the PCH, the onset can license extreme melodic complexity—such as prenasalization or glottalization—without violating the rhythmic requirement for a singular, low-intensity temporal beat. This explains why, in the diachrony, the C position serves as a magnet for features that eventually solidify into the primitive complex segments we observe in languages like Mazatec.

The Perceptual Compression Hypothesis (PCH) provides the necessary link between the physiological constraints of audition and the formal requirement for uncomplex onsets. By recognizing the onset as a point of maximal auditory responsiveness, we can understand why languages across the world prioritize a singular C slot. This position does not merely represent a structural boundary; it acts as a perceptual anchor that triggers the auditory system's sensitivity to the following rhyme.

Under the PCH, sequences that might otherwise violate the Sonority Sequencing Principle are diachronically reanalyzed. When the auditory system encounters a cluster at the onset edge, the heightened firing rate leads to temporal integration. The listener perceives a singular melodic bundle rather than a linear sequence of segments. This compression allows the phonology to maintain a strict CV template while still accommodating the dense melodic information found in languages like Mazatec.

Ultimately, the distinction between primitive and derived complex segments rests on the degree of this compression. Primitive segments are those that have been fully reanalyzed as

a single phonological unit—satisfying the criteria of being unpredictable, inseparable, and systemically symmetric. Derived segments, conversely, are emergent surface phenomena that occupy the C slot through active derivation but remain sensitive to morphological splitting and predictability. This theoretical framework now allows us to turn to the next section, where we will analyze the specific inventory of Huautla Mazatec, Sanskrit, and Mexican Spanish to determine which elements satisfy the requirements of primitive complexity and which remain surface-level derivations.

2.- Huautla Mazatec Complex Segments

In this section, I draw upon Wagner Oviedo (2026) regarding Huautla Mazatec to provide a foundational basis for the concept of primary complex segments. In this analysis of the language's extensive inventory of phonetic sequences—which includes typologically rare contrasts such as pre-aspirated versus post-aspirated consonants—I argue against a linear interpretation of these clusters. Instead, I propose that these sequences function as single complex phonemes whose internal arrangement is governed by a subsegmental hierarchical binary structure.

By adapting the "Immediate Constituents" model of Pike & Pike (1947) and integrating it into the framework of Q-Theory, I demonstrate that the contrast between these segments does not rely on temporal anchoring, but on a multi-layered architecture within a single segmental position. I adopt this subsegmental algorithm to define primary complex segments, using these findings to explain how such structures can host multiple phonetic elements while

maintaining the phonological behavior of a single unit. Under my proposal, these primary segments are distinguished from derived ones by their underlying hierarchical origin, providing the necessary empirical weight for the Perceptual Compression Hypothesis.

Below, I present the inventory of Huautla Mazatec consonants, followed by the justification for its analysis according to the parameters of predictability, separability, and distribution:

		Labial	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Glottal
Plosives	Plain	p	t				k	ʔ
	Pre-aspirated		^h t				^h k	
	Post-aspirated		t ^h				k ^h	
	Glottalised		tʔ				kʔ	
	Pre-nasalised	^m p	ⁿ t				^ŋ k	
	Pre-aspirated pre-nasalised		^{hn} t				^{hn} k	
	Pre-glottalised pre-nasalised	^{ʔm} p	^{ʔn} t				^{ʔŋ} k	
	Post-aspirated pre-nasalised		ⁿ t ^h				^ŋ k ^h	
	Post-glottalised pre-nasalised		ⁿ tʔ				^ŋ kʔ	
Affricates	Plain		^{ts}	^{tʃ}	^{tʂ}			
	Pre-aspirated		^h ts	^h tʃ	^h tʂ			
	Post-aspirated		ts ^h	tʃ ^h	tʂ ^h			
	Glottalised		tsʔ	tʃʔ	tʂʔ			
	Pre-nasalised		ⁿ ts	ⁿ tʃ	ⁿ tʂ			
	Pre-aspirated pre-nasalised		^{hn} ts	^{hn} tʃ	^{hn} tʂ			
	Pre-glottalised pre-nasalised		^{ʔn} ts	^{ʔn} tʃ	^{ʔn} tʂ			
	Post-aspirated pre-nasalised		ⁿ ts ^h	ⁿ tʃ ^h	ⁿ tʂ ^h			
	Post-glottalised pre-nasalised		ⁿ tsʔ	ⁿ tʃʔ	ⁿ tʂʔ			
Fricatives	Plain		s		ʂ			h
	Aspirated		s ^h		ʂ ^h			
	Glottalised		sʔ		ʂʔ			
Nasals	Plain	m	n			ɲ		
	Pre-aspirated	^h m	^h n			^h ɲ		
	Post-aspirated	m ^h	n ^h					
	Pre-glottalised	^ʔ m	^ʔ n			^ʔ ɲ		
	Post-glottalised	mʔ	nʔ			ɲʔ		
Laterals	Plain		l					
	Pre-glottalised		ʔl					
	Post-glottalised		lʔ					
Trills			r					
Taps			ɾ					
Approximants	Plain	β				j		
	Pre-aspirated	^h β				^h j		
	Post-aspirated	β ^h				j ^h		
	Pre-glottalised	^ʔ β				^ʔ j		
	Post-glottalised	βʔ				jʔ		

Based on the Perceptual Compression Hypothesis (PCH), the following analysis justifies the inventory of Huautla Mazatec by applying the parameters of predictability, separability, and distributional identification to define its complex sequences as primary complex segments.

Regarding predictability, I maintain that both voicing and the temporal anchoring of features are non-contrastive and governed by a subsegmental algorithm. In Huautla Mazatec, voicing is not an active phonological parameter; its surface realization is determined by the internal arrangement of subsegments. For instance, a nasal subsegment induces voicing in a following obstruent (e.g., /nt/ → [nd]), while glottal subsegments (h or ʔ) are responsible for maintaining voicelessness (e.g., /nth/). Furthermore, the phasing of these features follows universal phonetic tendencies where glottal features anchor to the right of onsets and nasality precedes the obstruent. By treating these as single temporal units, the PCH avoids the need for a phonological account based on contrasting timing, instead viewing the internal order as a predictable outcome of the segment's hierarchical structure.

The parameter of separability further reinforces this monophonemic analysis. Following Trubetzkoy's Rule I, these sequences cannot be segmented across syllabic boundaries because Huautla Mazatec strictly prohibits codas. This is evidenced by the total absence of word-final consonants, meaning any phonetic sequence must be contained within a single onset. Supporting this is the observation by G&K that these complex sequences appear within monomorphemic words (such as [n^hau¹] 'wind') and are resistant to segmentation by any morphological or morphophonological processes. Under the PCH, these are viewed as atomic units that undergo perceptual compression into a single segmental position rather than a string of independent phonemes.

Finally, distributional identification validates the proposal through the systemic symmetry of the language. In accordance with Trubetzkoy's Rule V, the consonant inventory exhibits a highly ordered array of intersecting contrasts—such as the correlations between plain plosives, pre-nasalized stops, and aspirated variants—which point toward a unified phonological grid. By adopting a Strict CV framework, I argue that the onset position admits these primary complex segments as single units. This perspective accounts for typologically rare features, such as word-initial pre-aspiration, by treating them as complex subsegmental architectures (e.g., (h n d) or (n t h)) rather than linear clusters, thereby providing a consistent explanation for the language's unique phonetic landscape.

Therefore, the derivation of a word such as [ʔndil] 'sterile' occurs in the following manner:

Subphonemic form: (t (n) ((?)))_C (i)_V

Phonemic form: (ʔ n d)_C (i)_V

Phonetic form: [ʔⁿd_C i^l_V]

It must be noted that within the phonemic form, following the proposal by Pike & Pike, there is a principal subsegment within the onset domain. This principal subsegment determines the phonological behavior of the complex segment in relation to phonological processes; for now, I will propose that the principal subsegment is the one with the lowest sonority in the case of C positions and the one with highest sonority in the case of V positions.

Huautla Mazatec also exhibits diphthongs and triphthongs. While certain diphthongs—such as /ao/, /ai/, /ia/, and /oa/—may appear as unpredictable primary forms, they are frequently found in derived environments. For example, the contrast between the monomorphemic [ʰtoa³⁴] ‘short’ and the morphologically complex [ʰkoa²] ‘my head’ (formed by ^hko ‘head’ + -a ‘1st person singular possessive’) demonstrates this. Triphthongs, on the other hand, are exclusively the result of derivation.

In these instances, I propose that a segmental merger occurs during the derivation process. This merger neutralizes the distinction between primary and derived complex segments at the phonemic level, rendering them indistinguishable on the surface. Within the PCH framework, these vocalic clusters are "compressed" into a single nucleus position, conforming to the Strict CV requirement while maintaining the internal phonetic complexity of the combined morphemes.

Subphonemic form:	(t (∅) ((h))) _C	(a (o)) _V	(t (∅) ((h))) _C	(o) _V	-(a) _V
Phonemic form:	(h t) _C	(o a) _V	(h k) _C	(o a) _V	
Phonetic form:	[ʰt _C oa ³⁴ _V]		[ʰk _C oa ² _V]		

In Huautla Mazatec, there are no phonological processes sensitive to syllable weight, nor does there appear to be a distinction between long and short vowels. Consequently, we can predict that any surface sequence within the boundaries of a CV unit constitutes part of a complex segment. However, it remains necessary to distinguish between primary and derived complex segments by applying the diagnostic parameters described above.

3.- Sanskrit Complex Segments

In applying the Perceptual Compression Hypothesis (PCH) to Sanskrit, the language's rich consonantal inventory can be categorized by distinguishing between segments that possess an inherent subsegmental architecture and those formed through phonological merger. I propose that the aspirated series, both voiced and voiceless (such as /b^h, d^h, g^h/ and /p^h, t^h, k^h/), should be analyzed as primary complex segments. Within the PCH framework, these are not treated as sequences of a stop plus a laryngeal fricative, but as single temporal units where the aspiration is a secondary feature anchored by a subsegmental algorithm. The timing of this aspiration is a predictable phonetic realization of the segment's internal hierarchical structure, and these units behave as atomic onsets that cannot be split by morpheme boundaries in their root forms.

Subphonemic form: (d (h))_C (u)_V

Phonemic form: (d h) (u)_vPhonetic form: [d^hu]

Gloss: 'agitate'

Conversely, I categorize sequences consisting of a fricative plus another consonant—such as /st, sp, ʃt/ or /rk/—as derived complex segments. Under a Strict CV framework, these sequences originate from separate underlying positions but undergo derivational fusion,

compressing them into a single segmental temporal position to satisfy surface onset requirements. Unlike the primary aspirates, these sequences are often separable, as I will show below.

Ultimately, the PCH proposes that at a specific point in the Sanskrit derivation, the distinction between a primary complex segment like /tʰ/ and a derived complex segment like /st/ is neutralized, with both occupying a single "segmental temporal unit" in the phonetic output. This classification places Sanskrit at the center of the proposed typological scale, maintaining a balanced distribution between inherent subsegmental complexity and derived cluster compression.

Van de Weijer (1996) discusses Sanskrit reduplication in the formation of perfect verb forms, noting several distinct behaviors regarding the onset. Aspirated initial consonants are reduplicated as their non-aspirated counterparts; for example, $d^h\bar{a}$ becomes $da-d^h\bar{a}$ ('to put/give').

Subphonemic form: (d (h))_C (a)_{VV}

Phonemic form: (d h)_C (a)_{VV}Reduplication: (d)_C (a)_V (**d h**)_C (**a**)_{VV}Phonetic form: [dad^ha:]

In the case of most consonant clusters, the reduplicant preserves the consonant with the lowest sonority, which is typically located on the left: *prat^h* becomes *pa-prat^h* ('to spread'), *kṣad* becomes *ca-kṣad* ('to divide'), or *svar* becomes *sa-svar* ('to sound').

While a positional rule could account for these cases, sequences of a fricative plus a plosive do not reduplicate the leftward consonant. Instead, they reduplicate the rightward consonant, which represents the segment of lowest sonority: *stu* becomes *tu-ʃtu* ('to praise') and *skand* becomes *ka-skand* ('to leap'). A combination of these criteria can predict the reduplicants of even more complex clusters, such as *sp^huʔ* becoming *pu-ʃp^huʔ* ('to burst') or *spɾd^h* becoming *pa-spɾd^h* ('to emulate').

Under the framework of the Perceptual Compression Hypothesis (PCH), I propose that these consonant clusters are treated as complex segments. Reduplication, therefore, operates by identifying the principal subsegment within the complex unit. This explains why only the principal subsegment—rather than the entire cluster—is present in the CV reduplicant.

Subphonemic form: (s)_C (t)_C (u)_V

Phonemic form: (s t)_C (u)_V

Reduplication: (t)_C (u)_V (s t)_C (u)_V

Phonetic form: [tustu]

Subphonemic form: (s)_C (p)_C (u)_V (t (h))_C

Phonemic form: (s p)_C (u)_V (t h)_C

Reduplication: (p)_C (u)_V (s p)_C (u)_V (t h)_C

Phonetic form: [pusput^h]

4.- Mexican Spanish Complex Segments

Mexican Spanish possesses only one primary complex segment: the phoneme /tʃ/. However, it features a rich inventory of diphthongs covering every combination of the five vowels, as well as the triphthongs /uau/ and /uei/. Additionally, sequences of obstruents plus liquids are largely unconstrained, appearing in a wide array of combinations such as /pl, tl, kl, pr, tr, kr, bl, gl, br, dr, gr/. The only notable exceptions are the prohibited sequence /dl/ and those involving strident sounds (/sl, sr, tʃl, tʃr/), which are invariably heterosyllabic, as seen in /is.la/ ‘island’ and /is.ra.el/ Israel. Under the principles of the Perceptual Compression Hypothesis (PCH), it is proposed that all these sequences—with the exception of /tʃ/—function as derived complex segments.

In this framework, /tʃ/ is categorized as a primary complex segment because its internal linear order (occlusion followed by friction) is an inherent, atomic property of its phonemic identity rather than a result of morphological combination or phonological rules. In contrast, the various vocalic and consonantal clusters in Mexican Spanish are analyzed as derived complex segments. Following the tenets of Strict CV theory, these sequences originate from separate underlying positions but undergo a process of segmental fusion to occupy a single temporal unit.

The diagnostic of separability is crucial here: while the components of /tʃ/ cannot be divided by syllable boundaries, groups like /sl/ or /tʃl/ are prohibited as onsets and must be split across syllables, indicating that the grammar evaluates the legitimacy of these combinations before allowing them to undergo "perceptual compression." Furthermore, the fact that most Spanish

diphthongs arise in derived environments—such as through cliticization or rapid speech—reinforces the classification of these structures as products of derivational fusion. Consequently, Mexican Spanish occupies a specific position on the PCH typological scale, characterized by a low density of primary complex segments and a high reliance on derivational processes to generate surface phonetic complexity.

To test the status of diphthongs as derived complex segments in Spanish under the principles of the Perceptual Compression Hypothesis (PCH), it is essential to analyze the behavior of reduplication in "Idioma Efe" (Wagner Oviedo, 2025). This language game functions as a diagnostic for separability and subsegmental hierarchy, revealing how the grammar processes the internal structure of the syllabic nucleus. In Idioma Efe, the process involves suffixing a reduplicant of the type $-fV$ to the right of the most sonorous element of the syllable, where the vowel of the reduplicant is a faithful copy of that element.

The data revealed by this process provides a pattern that strongly supports the PCH. In simple syllables like $/kasa/$, the result is $/kafasafa/$, where the reduplication is linear and direct. In rising diphthongs such as $/sielo/$, the result $/siefelofo/$ shows the reduplicant inserted after the most sonorous vowel ($/e/$), treating the sequence $/ie/$ as a unit where the head element dictates the copy. Similarly, in falling diphthongs like $/kausa/$, the process is triggered after the $/a/$, resulting in $/kafausafa/$ and leaving the glide $/u/$ outside the nucleus of the reduplicant. The most conclusive evidence comes from triphthongs, such as $/buei/ \rightarrow /buefei/$, where reduplication occurs immediately after the element of greatest sonority ($/e/$), ignoring the adjacent glides for the formation of the reduplicant.

From the perspective of the PCH, these data confirm that diphthongs and triphthongs in Spanish are derived complex segments. Although they present multiple vocalic elements on

the surface, the reduplication in Idioma Efe demonstrates that the grammar identifies a principal subsegment—the most sonorous element—to serve as the hierarchical head. This indicates that the reduplication process does not perceive a sequence of independent phonemes but rather a single complex segment. This reinforces the idea that, through segmental fusion, Spanish compresses vocalic sequences into a single nucleus position, maintaining an internal structure where only the dominant head is accessible for certain morphophonological processes.

5.- Summary

In summary, the application of the Perceptual Compression Hypothesis (PCH) across diverse linguistic systems demonstrates that phonetic complexity does not necessitate a departure from a streamlined phonological architecture. By distinguishing between primary complex segments, which possess an inherent subsegmental architecture, and derived complex segments, which result from the fusion of independent positions, this model reconciles surface variety with a universal structural template. This analysis relies heavily on the integration of Strict CV Theory, which posits that the phonological skeleton consists of an invariant succession of onset (C) and nucleus (V) positions. Under the PCH, complex onsets or diphthongs do not violate this "One-Slot" constraint; instead, multiple melodic elements are "compressed" into a single skeletal point. This skeletal uniformity is maintained while internal phonetic phasing is delegated to a subsegmental algorithm that manages the linear expression of features.

Furthermore, the PCH finds a formal home within the 3x Phonology framework developed by Scheer, which deconstructs phonology into three autonomous yet interacting tiers. The Skeletal Tier serves as the timing dimension, providing the slots where melody is anchored without carrying phonetic information. The Melodic Tier contains the actual features, such as place and manner, which in 3x Phonology are "object-oriented" and can be bundled into complex segments that satisfy the PCH. Finally, the Lateral Tier handles the relationships of Government and Licensing, explaining how a nucleus (V) can license a complex onset (C) to support its internal subsegmental burden. Unlike traditional branching onsets that require extra skeletal space, this lateral licensing allows a single C-slot to host multiple features as long as it is sanctioned by the following V.

By adopting the proposal that sonority is a phonological primitive, the PCH utilizes the lateral tier to explain hierarchical headedness. Sonority is not a gradient byproduct but a primitive force that determines which element in a "compressed" cluster acts as the principal subsegment. Whether it is the stop in a Sanskrit *s*-cluster or the most sonorous vowel in a Spanish triphthong, this head is the only part of the segment that phonological processes, such as reduplication, recognize and manipulate. Ultimately, this synthesis proves that human languages operate under a singular, sonority-driven logic that allows the strict constraints of the CV skeleton to accommodate phonetic diversity through the mechanism of perceptual compression.

In Huautla Mazatec, the "complexity" is largely lexical and primitive. The language utilizes the subsegmental algorithm to pack laryngeal information into the skeletal onset without needing to merge separate phonemes. Conversely, Mexican Spanish demonstrates derivational complexity. Its surface clusters look complex not because the phonemes

themselves are inherently "packed," but because the grammar's lateral tier is highly permissive, allowing separate melodic units to undergo segmental fusion into a single skeletal position.

Sanskrit occupies the middle ground of this scale, utilizing both primary aspiration and derived fricative-plosive clusters. In all three cases, the Perceptual Compression Hypothesis maintains the integrity of the Strict CV skeleton while using sonority as the primitive guide to organize the internal hierarchy of the resulting complex segments.

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